



MRSPlotter

User manual and documentation

Version 1.0

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Developed by: Lili Fanni Tóth at MIDAlab, Institute of Biotechnology and Biomedicine (IBB), Universitat Autònoma de Barcelona (UAB)

Overview:

MRSPlotter is a specialized software tool designed for visualizing magnetic resonance spectroscopy data and generating publication-quality figures from various file formats. The application accommodates both single-voxel and multivoxel spectral data files. Users can view and edit the data and subsequently export it in CSV or standardized XML format. This workflow enables modifications, such as adding missing information, before using the files for plotting. MRSPlotter produces a range of plot types, including subplots and overlaid spectra, and exports figures suitable for both publication and presentation.

The software is open source, with its source code freely accessible under the CC BY-NC-SA 4.0 Plus license on GitHub (<https://gitfront.io/r/lilively/Mi476tpG8Eq5/MRSPlotter/>). A compiled executable version is available for users who prefer direct installation. For users compiling from source, all necessary libraries are specified in the requirements text file included in the repository.

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1. Introduction

jMRUI2XML^{1,2} generates XML files in a standardized format. These files encapsulate both spectral data and essential metadata. However, once the spectra are processed, they cannot be reopened with jMRUI3. XML files can be opened using Notepad++ or Excel. Manually extracting intensity values and reconstructing frequency values in ppm—since only the first and last ppm are included in the XML file—is tedious, especially with tens or hundreds of spectra.

On the other hand, jMRUI-generated text files do not directly export frequency values in ppm. Instead, the frequency scale can be reconstructed from acquisition parameters such as the sampling interval, transmitter frequency, and reference frequency, using the Nyquist theorem, to achieve precise spectral visualization.

Visualizing processed spectra is crucial for quickly assessing spectral quality and identifying key metabolic patterns across different tissue types. To address this challenge, MRSPlotter was developed (Figure 1). It integrates a graphical user interface with Python scripts. This tool streamlines the workflow from data acquisition to the creation of publication-ready figures. It enables simultaneous processing of hundreds of spectra without requiring programming expertise.

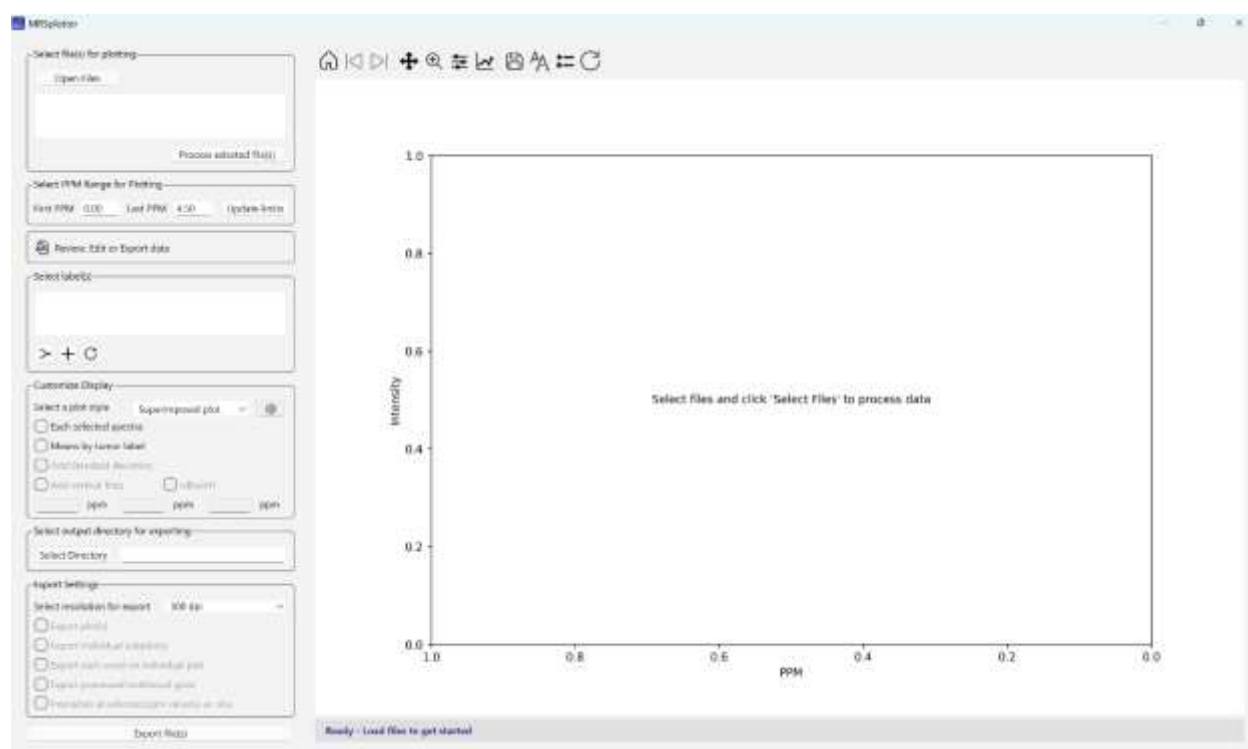


Figure 1 Main interface of MRSPlotter.

Once loaded, the data can be explored and modified within the application or exported for further adjustments offline.

The application supports both superimposed plots (multiple spectra displayed together for direct comparison) and subplots (plots for each tissue type) for individual-tissue analysis. Additionally, it

allows users to define custom classes by combining up to four different tissue types, facilitating tailored analyses across datasets.

The default ppm range is 0 to 4.5, but it can be easily changed by inputting a desired ppm range. Plot options include displaying mean spectra with or without standard deviations, either individually or overlaid on the spectra of all cases. The toolbar provides additional customization options, including adjusting font size, modifying the layout, and editing axis and curve parameters. The displayed figure can be exported at 600 dpi directly from the toolbar or through the export options when an output directory is selected.

For quick alignment and quality control, up to 3 vertical reference lines can be added to the plot. For example, if “isBrainH” is selected, lines can be placed at 1.29, 3.03, and 3.21 ppm. The user can choose up to three ppm values to display. Additionally, the intensity values at the specified ppm points can be exported as an Excel file, including case IDs and corresponding labels. This facilitates statistical analysis of peak heights or ratios.

MRSPlotter provides several interactive plotting options, such as:

- Interactive options to preview and edit the data
- Interactive color selection for tissue types
- Dynamic plot navigation with zoom/pan capabilities
- Dynamic adjustments of PPM ranges and display options
- Export files for offline modifications and further analysis

2. Requirements

- Operating System: Windows 10 or 11 (may work on Windows 7/8)
- RAM: Minimum 4GB recommended (8GB+ for larger datasets)
- Disk Space: At least 200MB free space
- Permissions: Write access to folders where you intend to save exported plots

For Python installation:

- PyQt6 >= 6.4.0
- matplotlib >= 3.5.0
- numpy >= 1.21.0
- pandas >= 1.3.0
- regex >= 2022.1.18

The dependencies can be installed from the requirements text file with the following command:

```
pip install -r requirements.txt
```

3. Installation

3.1 Windows installation (Standalone Executable)

After downloading the latest version from the GitHub releases page, extract the contents and start the application by double-clicking on the MRSPlotter.exe.

3.2 Source Code Distribution

Full source code can be found on GitHub. To get it, clone the repository to your local machine:

```
git clone https://github.com/lilively/MRSPlotter
```

Navigate to the project directory and run the main script:

```
python main.py
```

4. Supported formats

- Single voxel files created with jMRUI2XML^{1,2}
- Multi-voxel files generated with jMRUI2XML^{1,2}
- Datasets created with Spectra Classifier³
- Single voxel jMRUI Data Text file created with jMRUI version 6 or 7⁴
- Multi-voxel jMRUI Data Text file created with jMRUI version 6 or 7⁴
- Comma-separated values (CSV) files exported from the application
- Input data matrix exported from MRSdecomposer

5. Loading data

1. Click on the “Open Files” button to select one or more XML files
2. Select the files you would like to plot by clicking on them
 - “Ctrl + A” shortcut: select all files
 - “Ctrl + D” shortcut: deselect all files
2. Click on “Process selected file(s)” for plotting to load the files
3. Upon successful loading, the labels found will be found in the “Select label(s)” list box. Each spectra will be selected, and the first label found will be displayed
4. Select as desired from the available plotting sections

5.1 JMRUI Data Text file

In case of obtaining the XML with JMRUI2XML is not possible, the jmrui text format (Figure 2) may be available. In the header of the file, there are various parameters, like “PointsInDataset”, “TransmitterFrequency”. If multiple cases are in the file, the “DatasetsInFile” field indicates the number, and the file names are stored in the “SignalNames” field.

```

JMRUI Data Textfile

Filename: se-batch4.txt

PointsInDataset: 512
DatasetsInFile: 6
SamplingInterval: 1E0
ZeroOrderPhase: 0E0
BeginTime: 0E0
TransmitterFrequency: 6.3903E7
MagneticField: 1E0
TypeOfNucleus: -1E0
NameOfPatient:
DateOfExperiment:
Spectrometer:
AdditionalInfo:
SignalNames: SV_375-SE-W.mru1;SV_382-SE-W.mru1;SV_389-SE-W.mru1;SV_391-SE-W.mru1;SV_394-SE-W.mru1;SV_396-SE-W.mru1;

Signal and FFT
sig(real)      sig(imag)      fft(real)      fft(imag)
Signal 1 out of 6 in file
-1.0659E-2      -3.9063E-3      5.7613E-3      2.0721E-3
-2.0954E-2      -7.5614E-3      5.8248E-3      2.0102E-3
-1.9372E-2      -7.6412E-3      5.745E-3       1.8855E-3
-1.8881E-2      -8.2037E-3      5.8688E-3      1.8335E-3

```

Figure 2 JMRUI text format example.

5.1.1 Generation of frequency axis

The frequency axis in ppm is calculated by determining the bandwidth in Hz from the sampling interval, then applying the Nyquist theorem to find the spectral width.

Hz per point is calculated by dividing this spectral width by half the number of points. For each data point, the frequency is centered around zero by subtracting half the number of points from the index, multiplying by Hz per point to obtain the frequency offset, then adding the reference frequency. Finally, conversion from Hz to ppm is achieved by dividing each frequency by the transmitter frequency (in MHz) multiplied by 10^6 .

Unfortunately, the reference frequency is not stored in the jmrui text file, so the frequency axis is incorrect (Figure 3).

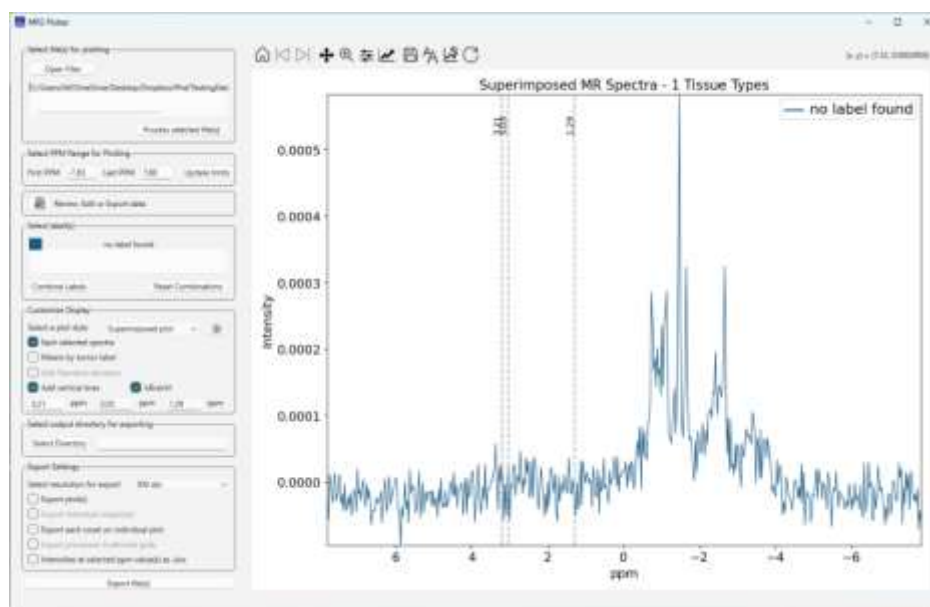


Figure 3 Jmruif datafile plotted without reference frequency at 0

A workaround is clicking on “File, Print/Export” in jmrui. When the window pops up (Figure 4) the necessary reference frequency value line can be copied to the text file (Figure 5). When having this additional value, the frequency axis is corrected (Figure 6). Centering around 4.75 ppm is also an option; however, it does not work for all cases, and the small shifts in the alignment are more difficult to recognize.

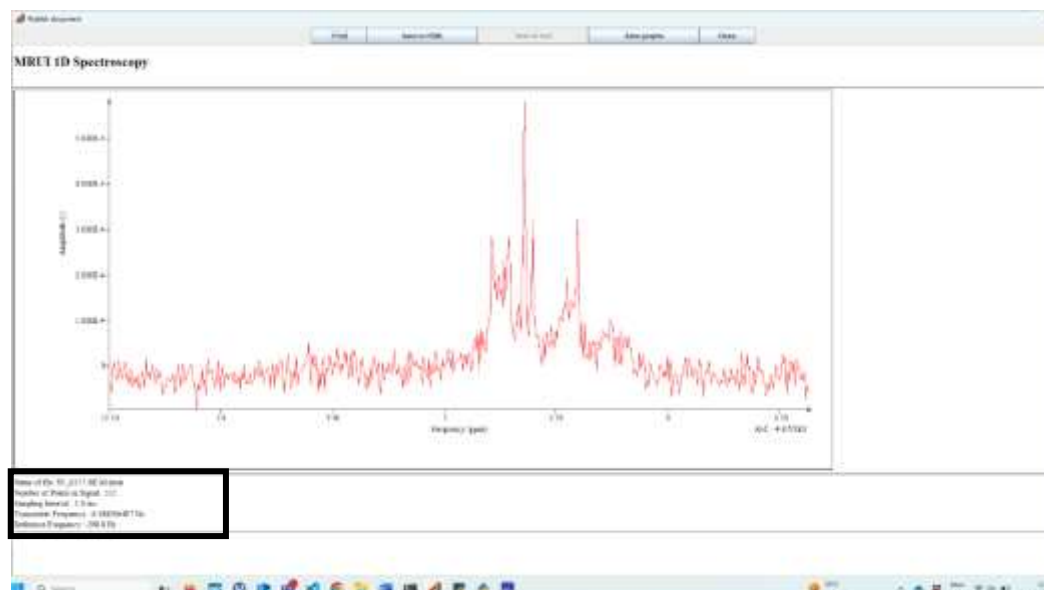


Figure 4 jMRUI Print/Export window

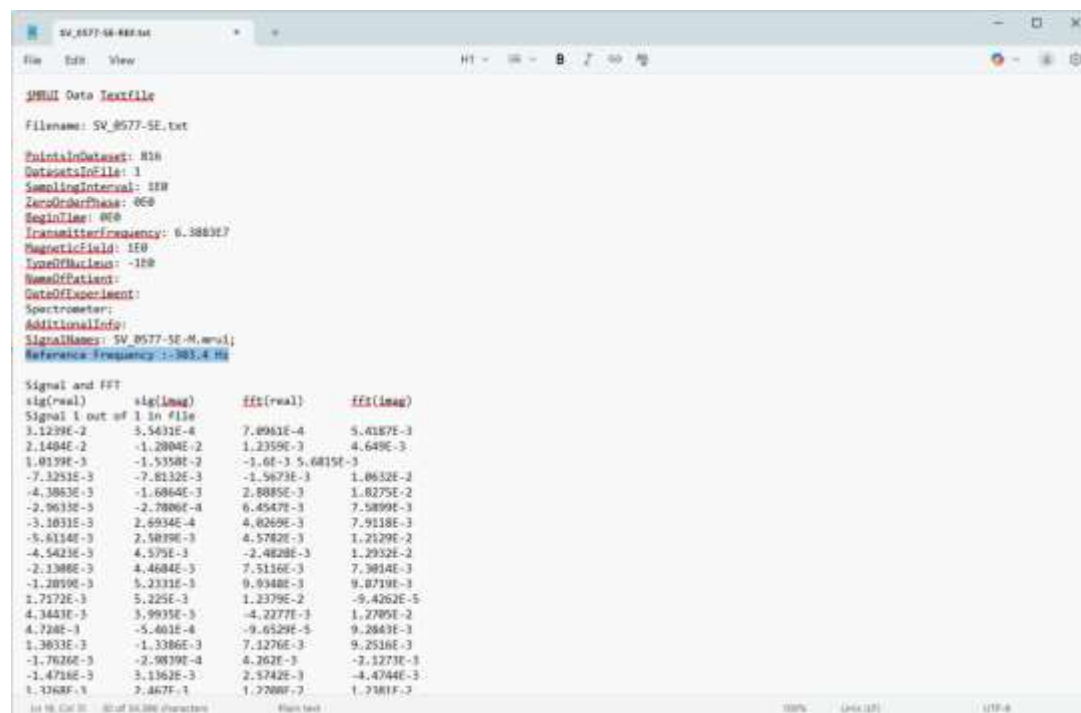


Figure 5 jMRUI text file with added reference frequency

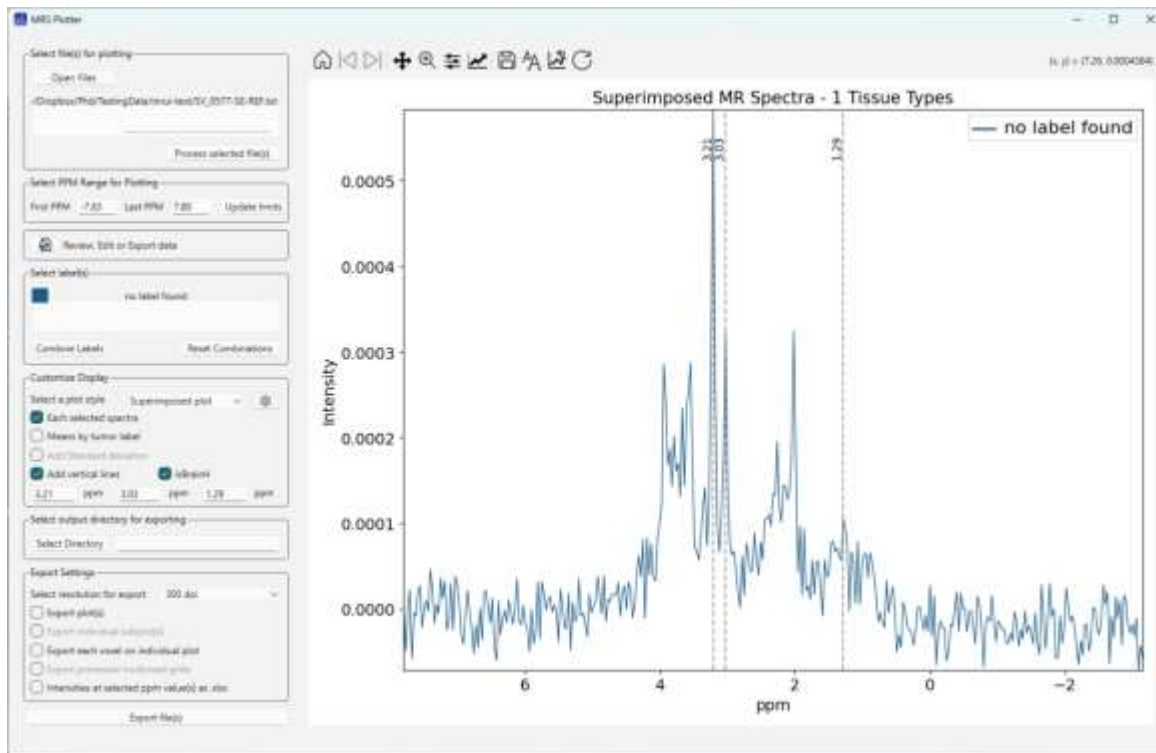


Figure 6 Corrected frequency axis

6. Select ppm range

The default first ppm is 0.0, and the last ppm is 4.5. In case the files have either the first ppm or last ppm out of bounds, a warning – like Figure 7 – is displayed.

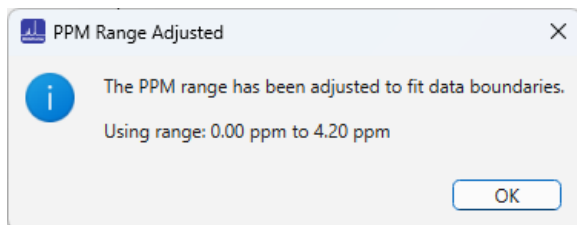


Figure 7. Pop-up window, warning that the ppm range is adjusted

The ppm range can be modified at any time, but reloading the data will lose all previous changes.

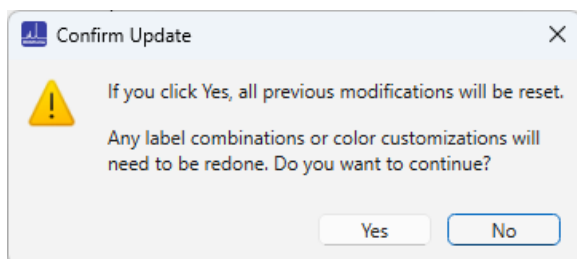


Figure 8. A pop-up window warns that adjusting the ppm range will reload the data and reset previous settings.

7. Review, edit, and export data

The displayed data can be previewed by clicking on the file icon. A confirmation window appears (Figure 9) to verify the action. This may take a while for large files.

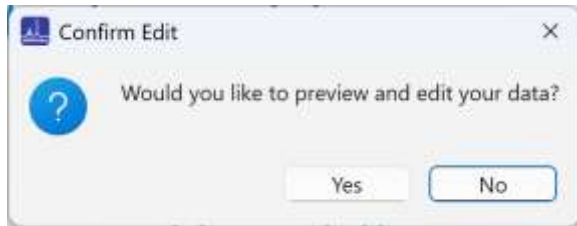


Figure 9. Confirm opening the data edit window.

Once confirmed, the data editor window opens (Figure 10). The window provides a comprehensive interface for viewing, filtering, and modifying the data.

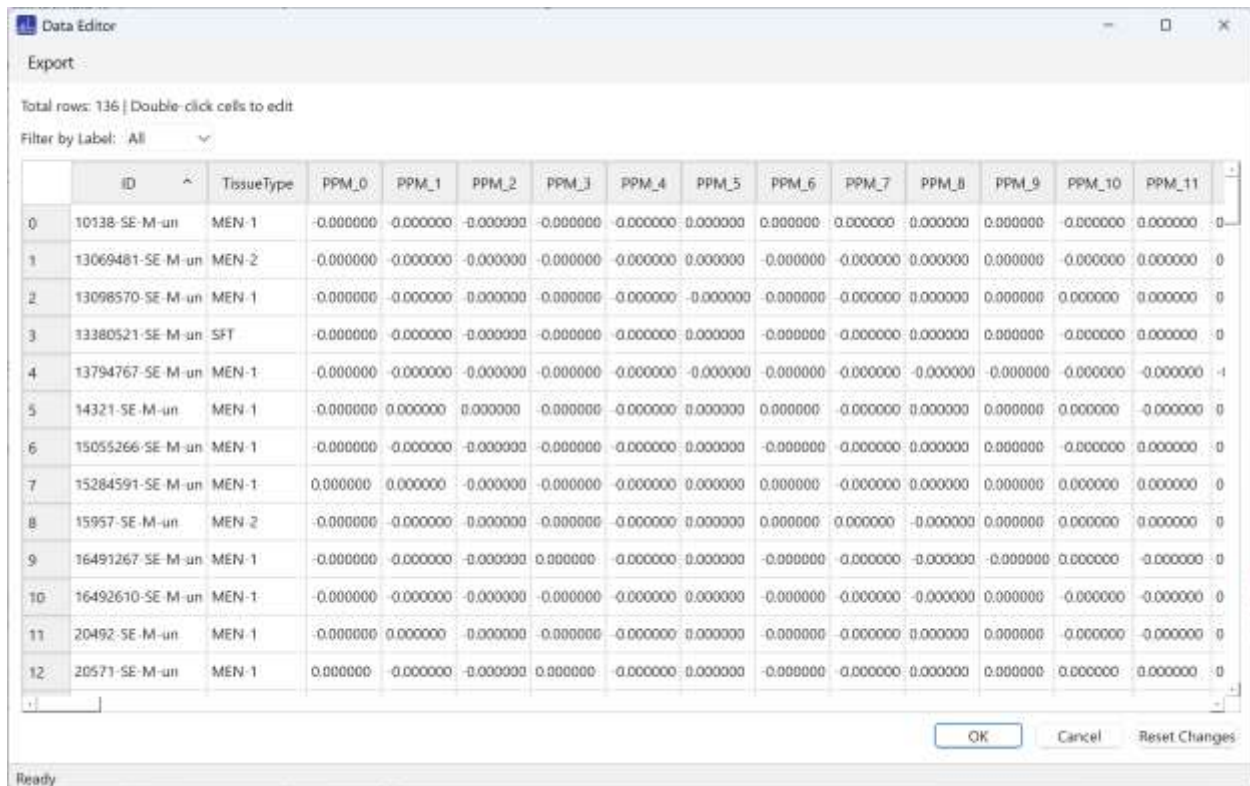


Figure 10 Data editor window.

Below the menu bar is a filtering section that allows users to filter data by tissue type labels. The dropdown menu labeled "Filter by Label:" offers the option to view "All" data or select a specific tissue type from those present in the dataset. When a particular type of tissue is selected, only the rows with matching tissue type are displayed in the table, and the status bar updates to show the number of filtered rows (Figure 11)

Data Editor

Export

Total rows: 136 | Double-click cells to edit

Filter by Label: SFT

	ID	TissueType	PPM_0	PPM_1	PPM_2	PPM_3	PPM_4	PPM_5	PPM_6	PPM_7	PPM_8	PPM_9	PPM_10	PPM_11	PPM
0	13380521-SE-M-un	SFT	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	0.000000	-0.000000	-0.000000	0.000000	0.000000	-0.000000	0.000000	0.000000
1	121022-SE-M-un	SFT	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	0.000000	-0.000000	-0.000000	0.000000	0.000000
2	15040645-SE-M-un	SFT	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	0.000000	0.000000	-0.000000	0.000000
3	15222877-SE-M-un	SFT	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	0.000000	-0.000000	-0.000000	0.000000	-0.000000	-0.000000	0.000000
4	63683-SE-M-un	SFT	-0.000000	-0.000000	-0.000000	0.000000	-0.000000	-0.000000	0.000000	0.000000	-0.000000	-0.000000	0.000000	0.000000	-0.000000

OK Cancel Reset Changes

Filtered: showing 5 of 136 rows

Figure 11 Example for filtering.

The rest of the window is occupied by a sortable data table that presents the following information:

- "ID" column: Identifying information for each spectra
- "TissueType" column: Classification of the tissue type for each spectra
- PPM value columns: The spectral data filtered to the selected PPM range

Users can interact with the table in several ways:

- Double-click on cells to edit values directly
- Sort data by clicking on column headers
- Select multiple cells/rows for copying data

Keyboard shortcuts can be used for common operations:

- Ctrl+C: Copy selected cells
- Ctrl+V: Paste data from the clipboard
- Ctrl+A: Select all cells
- F11: Toggle full screen mode

The window also features a menu bar with an "Export" option that allows users to save the current data selection (Figure 12)

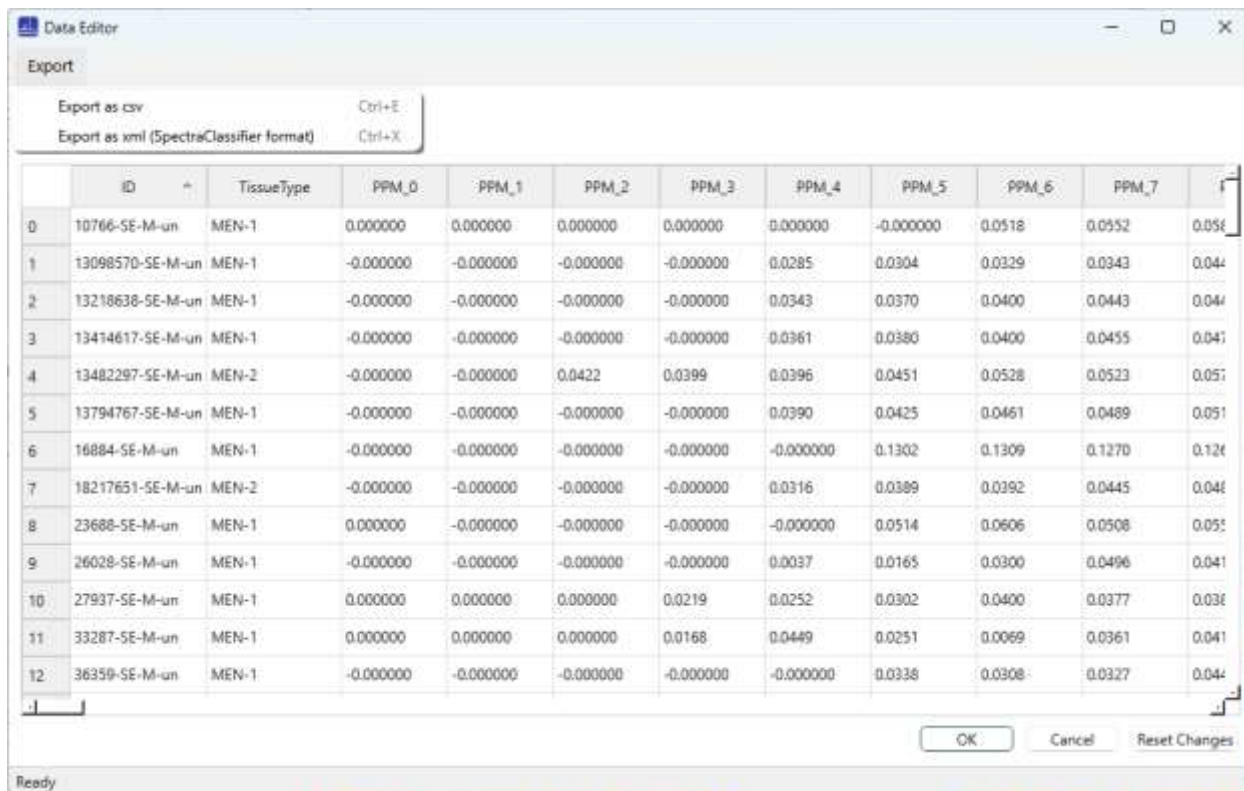


Figure 12 Export options.

The first option is export as a CSV file, which can be opened with other applications, like simple text editors, to edit (Figure 13). The other export format follows the SpectraClassifier data format, which is the same for single- and multivoxel data. This means that the xml files can be seamlessly integrated into the existing analysis pipeline.

At the bottom of the window, users can confirm changes by clicking the "OK" button, which saves all modifications and returns to the main interface. Alternatively, the "Cancel" button closes the window without applying any changes, preserving the original data. If users attempt to close the window with unsaved changes, a confirmation dialog appears, asking whether they want to save their changes before closing, preventing accidental data loss.

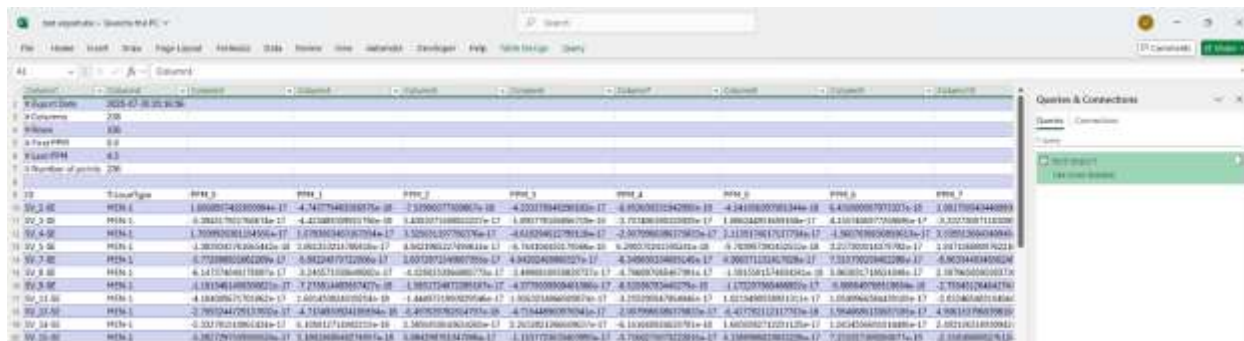


Figure 13 Exported SFT data opened with excel.

Changing the data labeling gives users flexibility; for example, Figure 14 shows when the label is changed from the tissue type to the case ID.

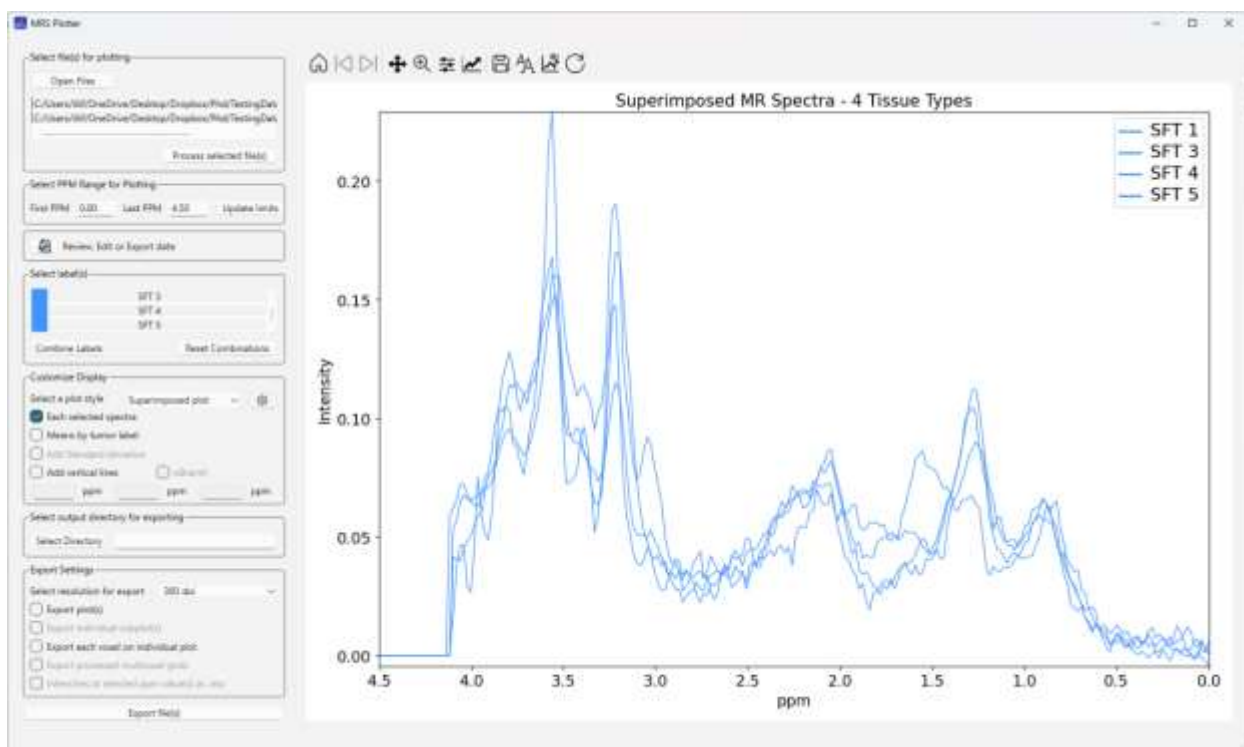


Figure 14 Unique SFT cases.

8. Label(s)

Unique classes from the XML files will be loaded into a list box and displayed in alphabetical order. Initially, random colors are assigned to each class. By double-clicking a class, a color picker menu (Figure 15) appears, allowing a specific color to be assigned. Custom colors can be created by clicking one of the white squares, selecting a color, and clicking “add to custom color”. The generated colors are saved, allowing us to reuse them after closing and reopening the application. Colors can also be input using hexadecimal codes, making it easy to replicate previous plots. Additionally, shortcuts for selecting all or deselecting are available, as described earlier.

Figure 15. Color picker menu for assigning label colors.

8.1 Combine labels

Custom labels can be created by creating classes from the labels available (Figure 16). Up to 4 classes can be created at the same time, but clicking “Combine Labels” creates new combinations. Clicking “Reset Combinations” (Figure 17) reloads the original labels with randomly assigned colors.

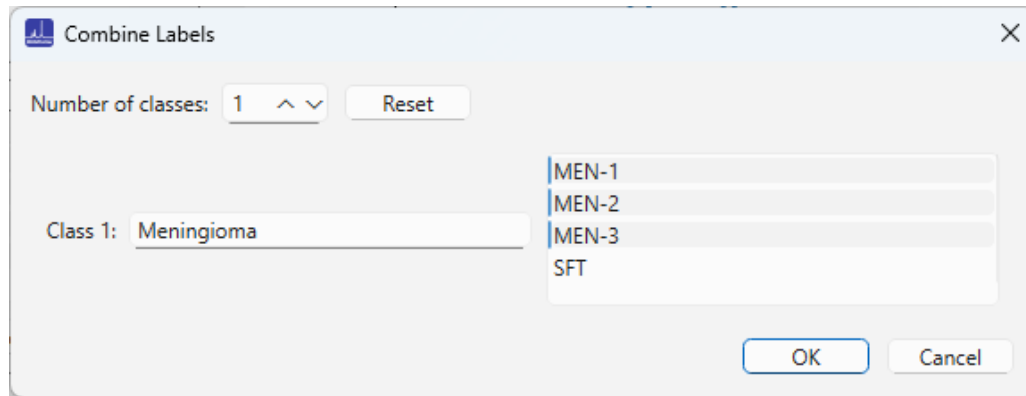


Figure 16. A pop-up window that allows the creation of classes from labels.

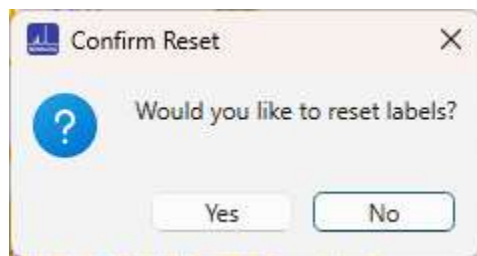


Figure 17 Pop-up window for confirming resetting the labels

8.2 Importing label(s) from MRSdecomposer results

The MRSdecomposer output files contain, for each file, X and Y positional coordinates and the winning source contribution. By clicking the “Load labels” option, this file can be used to assign labels to the input file. If multiple repetitions are detected, a pop-up window will prompt the user to select one of them.

9. Plot layout types

9.1 Superimposed plots

All selected cases are shown on a single plot, with colors indicating their labels.

9.2 Subplots

Each label is rendered on its own individual plot within an automatically configured grid layout. The grid dimensions can be customized using the settings button (gear icon).

9.3 Multivoxel grid

Available only for multivoxel cases. Displays the processed spectra for all voxels using the original grid dimensions from the source file. When multiple files are selected, only the first case is displayed by default, though additional cases can be exported for analysis.

9.4 Display Configuration

9.4.1 Each selected spectra

When selected, the spectra for each case are shown as on Figure 18.

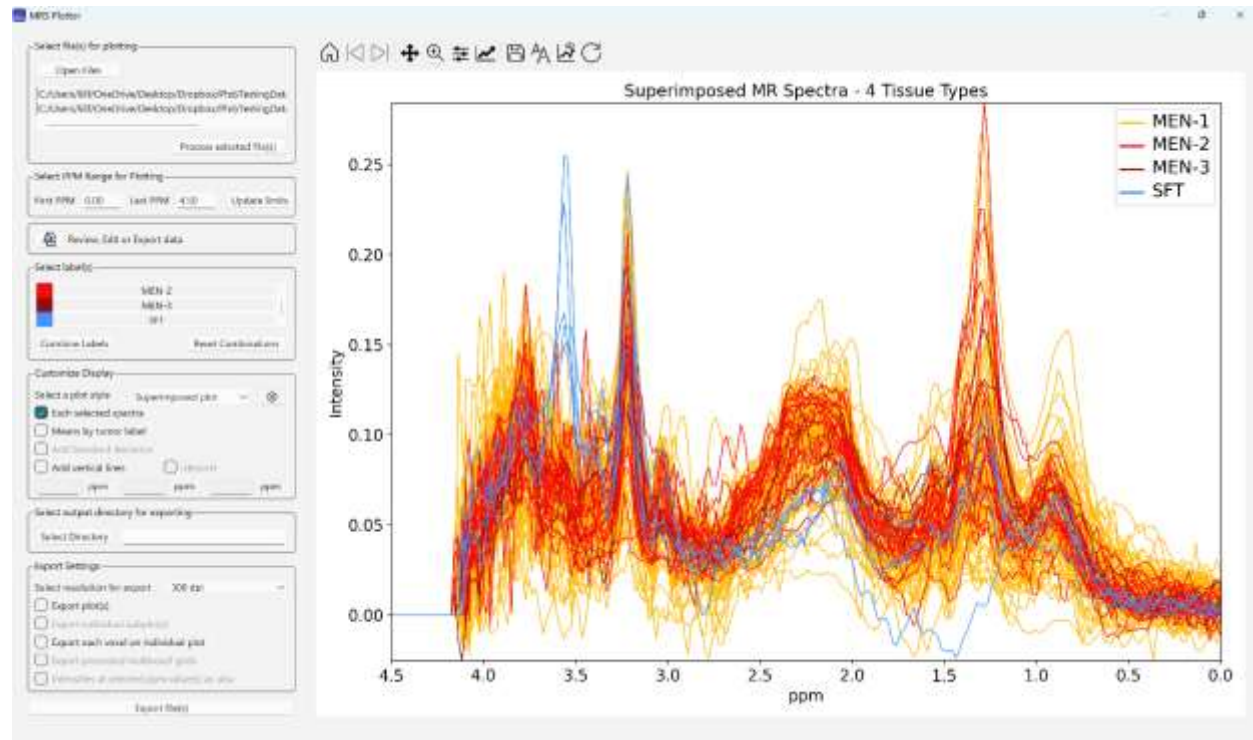


Figure 18. Each selected spectra of the meningioma dataset (short echo) is shown on a superimposed plot.

9.4.2 Means by tumor label

When selected, the mean spectra of each class are shown. When the “Each selected spectra” is also selected, the mean of the given label is plotted with a darker and thicker version of the chosen color to avoid confusion.

9.4.3 Add standard deviation

Available when “Means by tumor label” is selected. (Figure 19)

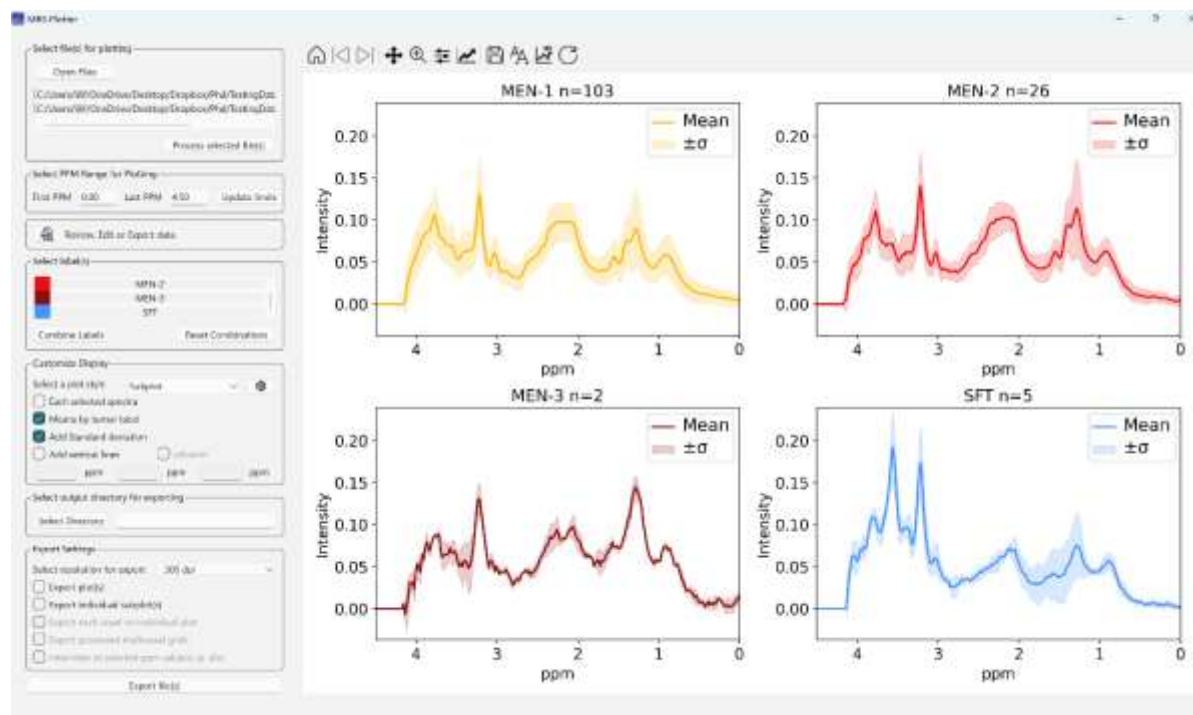


Figure 19. Mean and standard deviation of the meningioma dataset (short echo) on subplots.

9.4.4 Add vertical lines

This option includes reference lines at key metabolite positions to facilitate peak assessments. When "isBrainH" is selected, the fields are automatically populated with 1.29, 3.03, and 3.21 ppm. Up to three values can be used at the same time (Figure 20).

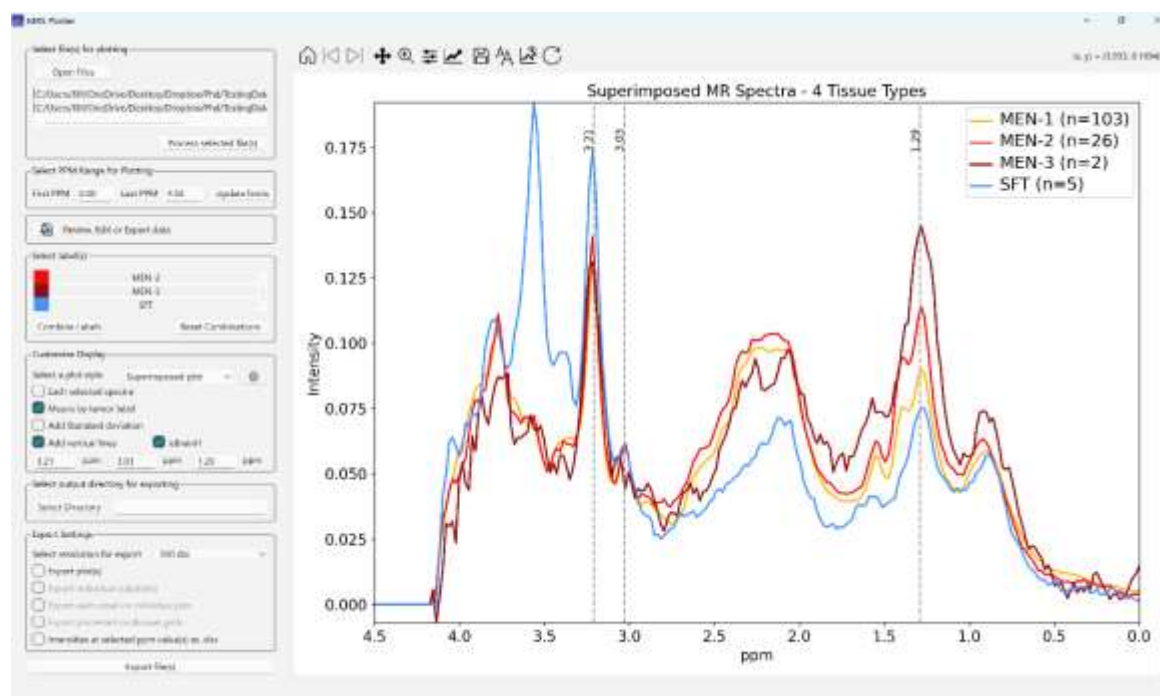


Figure 20. Mean spectra of meningiomas with vertical line added at 3.2, 3.03 and 1.29 ppm.

9.4.5 Configure Grid size for subplots

The default subplot configuration uses up to three columns, with additional rows added as needed to accommodate all plots. Users can modify the grid dimensions by clicking the gear icon, which opens the grid configuration dialog. (Figure 21). By unchecking the automatically determined grid layout option, the row and column values can be easily modified. This flexibility allows for optimal layout arrangement based on the number of spectra and display preferences. (Figure 22)

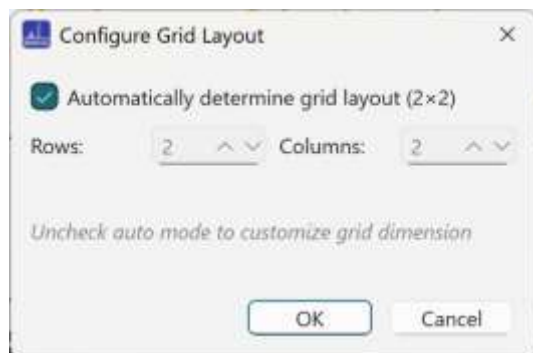


Figure 21 Configure grid window.



Figure 22 Short echo spectra of SFTs.

10. Toolbar options

Interactive navigation is achieved by using matplotlib's interactive toolbar, with two additional buttons for changing font size and resetting the plot to its original state. (Figure 23)



Figure 23. Toolbar for customizing displayed plot

10.1 Home

Resets zoom and panning to the default view.

10.2 Forward & Back

Navigate between zoom steps. Only available when zoom modifications have been applied to the plot

10.3 Zoom

Allows zooming into a selected rectangular region of the plot, including the legend.

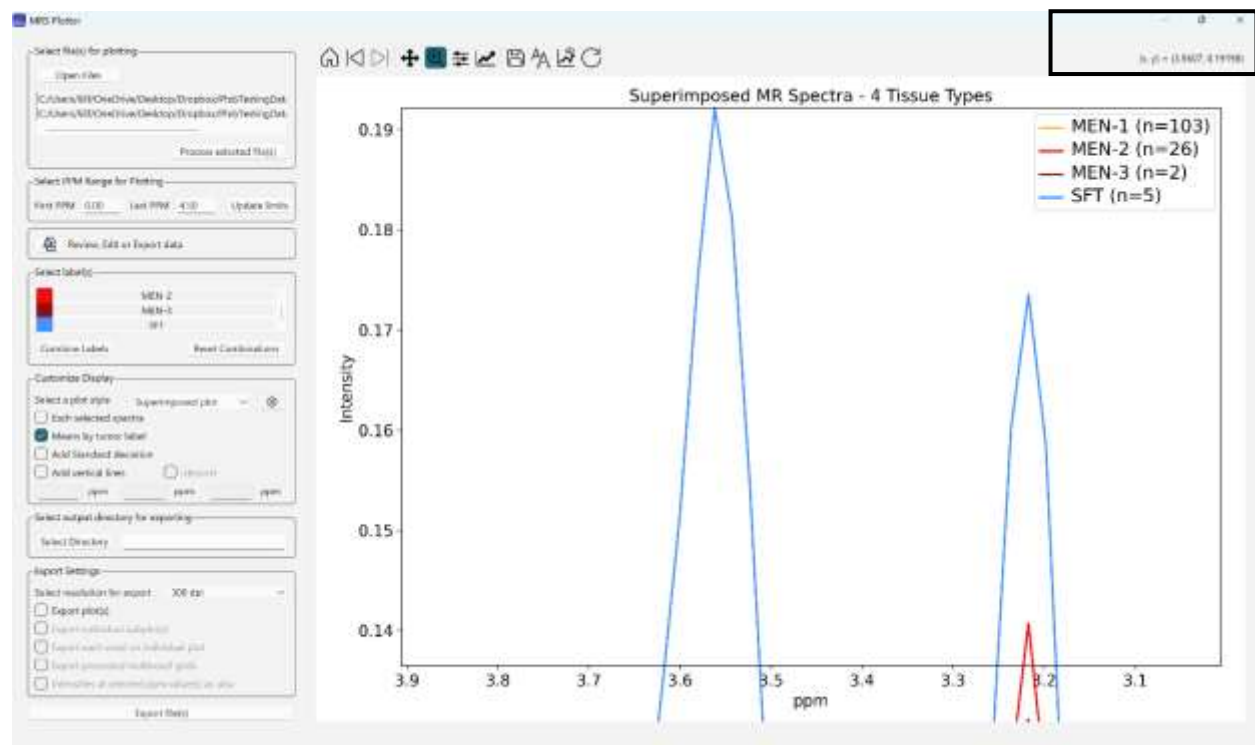


Figure 24. When zooming, the peak's intensity and frequency can be easily checked, as the x and y positions of the curve are displayed on the top right (purple square).

10.4 Configure subplots

The positions of the axes and border margins can be modified to achieve the desired figure layout. This adjustment capability is particularly beneficial when using custom grid dimensions instead of the automatic grid selection (Figure 25)

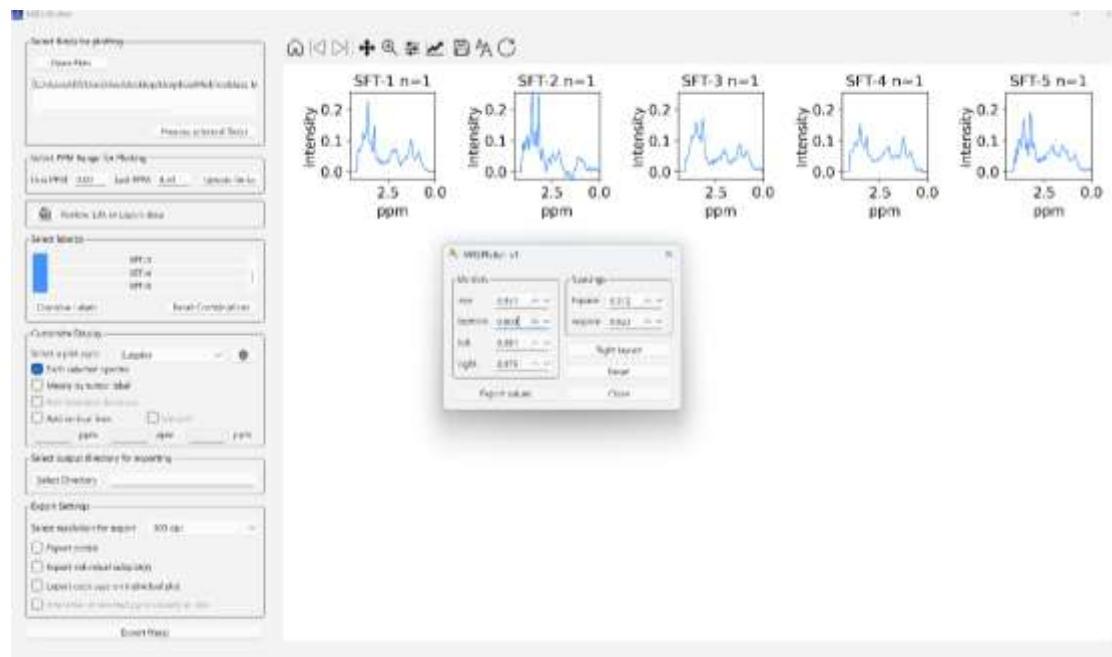


Figure 25. Optimizing layout for subplot.

10.5 Edit Axis, Curve, and Image Parameters

Customize individual plot elements, including axis labels, plot titles, line styles, line widths, marker sizes, and legend labels. After modifying legend labels, click "Regenerate Automatic Legend" to apply the changes to the legend shown. (Figure 26)

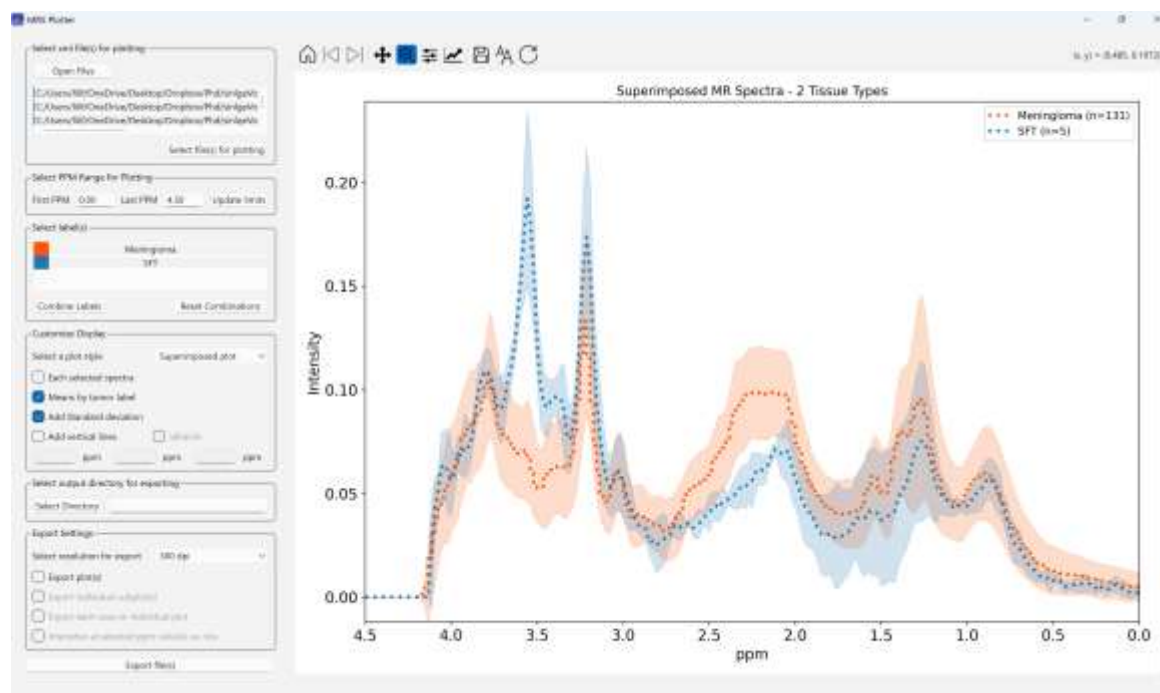


Figure 26. Line style changed to dotted. The legends are updated when "Regenerate Automatic Legend" is selected.

10.6 Save

Exports the plot as displayed in various formats, including JPEG, PNG, PDF, TIFF, and EPS. The resolution is set to 600 dpi.

10.7 Change Font Size

Apply the selected font size to all text elements in the display. Note that plot layout adjustments may be necessary after font size changes to maintain optimal spacing and readability

10.8 Change legend visibility

Toggle legend display on or off for all plots in the current view. This control simultaneously affects legend visibility across the entire display.

10.9 Reset Figure

Restore the figure to its original default state, removing all customizations and modifications. This action cannot be undone.

11. Output directory

The output directory must be specified if any export options are selected.

12. Export settings

Plots generated can be exported in 300 (default), 600 or 800 dpi.

12.1 Export plot(s)

Exports a displayed plot or subplots with predetermined settings. The filename generated contains information about the selected plotting options.

12.2 Export individual subplot(s)

Available if the plot style is set at “Subplot”. Each subplot is exported with predetermined settings.

12.3 Export each case on an individual plot

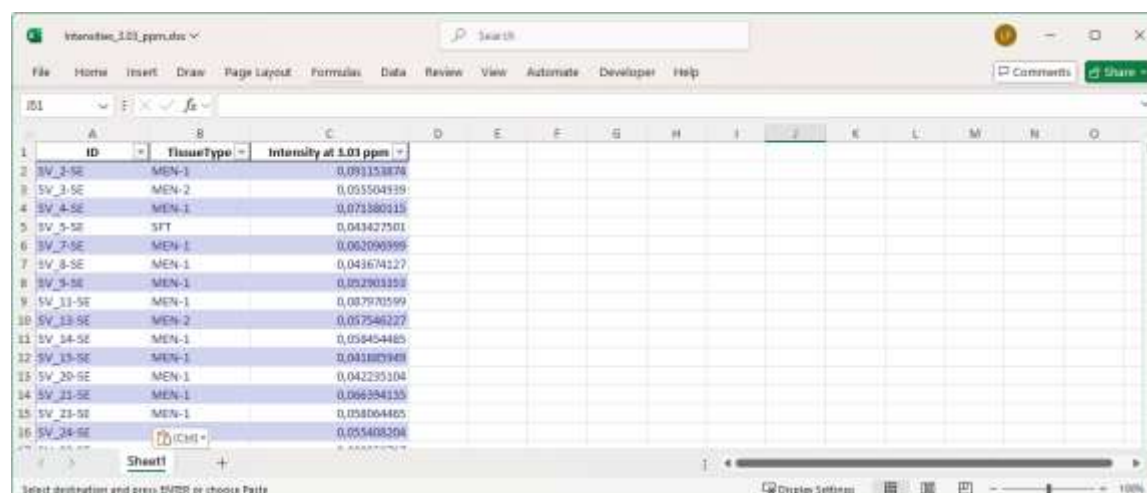
Available only when “Each selected spectra” is selected. Creates a plot for each case. As this can mean many plots, this is happening in the background; the plots are not displayed but directly exported. The currently exporting filename is shown in the status bar at the bottom of the application.

12.4 Export multivoxel grids

Available only when the plot style is “multivoxel grid”. Creates multivoxel grid plots for each case. As this can mean a large number of plots, this is happening in the background, the plots are not displayed but directly exported. The currently exporting filename is shown in the status bar at the bottom of the application.

12.5 Intensities at selected ppm value(s) as xlsx

Available only when “Add vertical lines” is selected. When at least one box contains a ppm value, the intensities of the selected cases are exported to the output directory (Figure 27).



ID	TissueType	Intensity at 1.01 ppm
SV_3-SE	MEN-1	0.091158878
SV_3-SE	MEN-2	0.055504919
SV_4-SE	MEN-1	0.071380115
SV_5-SE	SFT	0.043427501
SV_7-SE	MEN-1	0.062090999
SV_8-SE	MEN-1	0.043674127
SV_9-SE	MEN-1	0.052903153
SV_11-SE	MEN-1	0.007970599
SV_13-SE	MEN-2	0.057546227
SV_14-SE	MEN-1	0.058454465
SV_15-SE	MEN-1	0.041803949
SV_20-SE	MEN-1	0.042235104
SV_21-SE	MEN-1	0.066394135
SV_23-SE	MEN-1	0.058064465
SV_24-SE	MEN-1	0.055408304

Figure 27. Example of excel output.

13. Examples

13.1 Interpret dataset⁵

When the selection contains more than 10 tissue types, the generated labels are smaller to fit better (Figure 28).

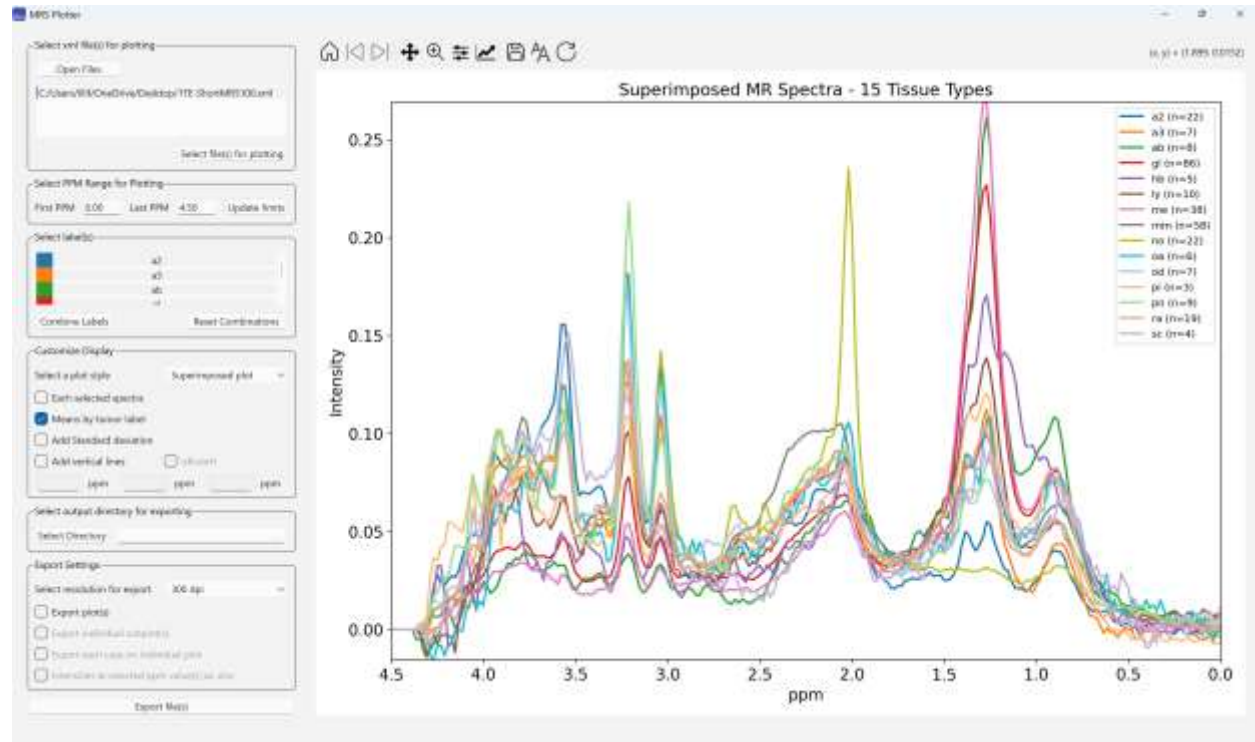


Figure 28. Mean of interpret short echo single voxel dataset (15 tissue types).

13.2 Example: Multi voxel dataset⁶

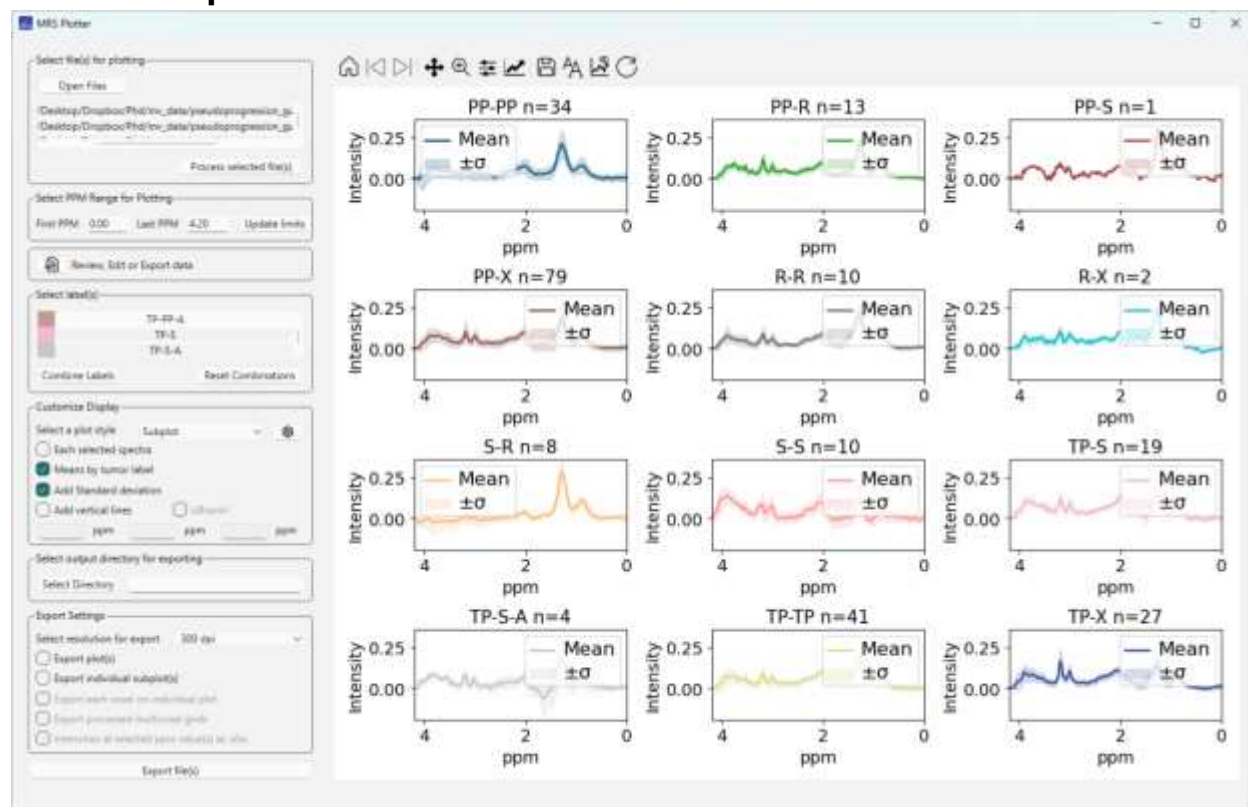


Figure 29. Labels from the pseudo-progression multi-voxel dataset.

As it can be seen on Figure 29 When many labels are selected, individual plots are quite squished, and the legends overlap. The labels can be removed from the individual figures or with all at once with the toolbar (Figure 30), however, when exporting the image looks proportional (Figure 31).

Figure 32 shows what an exported voxel from a multivoxel case looks like. The x and y positions are shown in the title and in the filename.

Handling numerous spectra, such as from a multivoxel dataset, can be time-consuming as the application takes a while to respond (Figure 33). Since the plots are regenerated with each change, it's advisable to exercise caution when plotting thousands of spectra.

Figure 34 shows that when “Multivoxel grid ” is selected only the first case is shown, but as been seen on the status bar, in the background all processed cases are exported. Figure 35 shows one these exported grids.

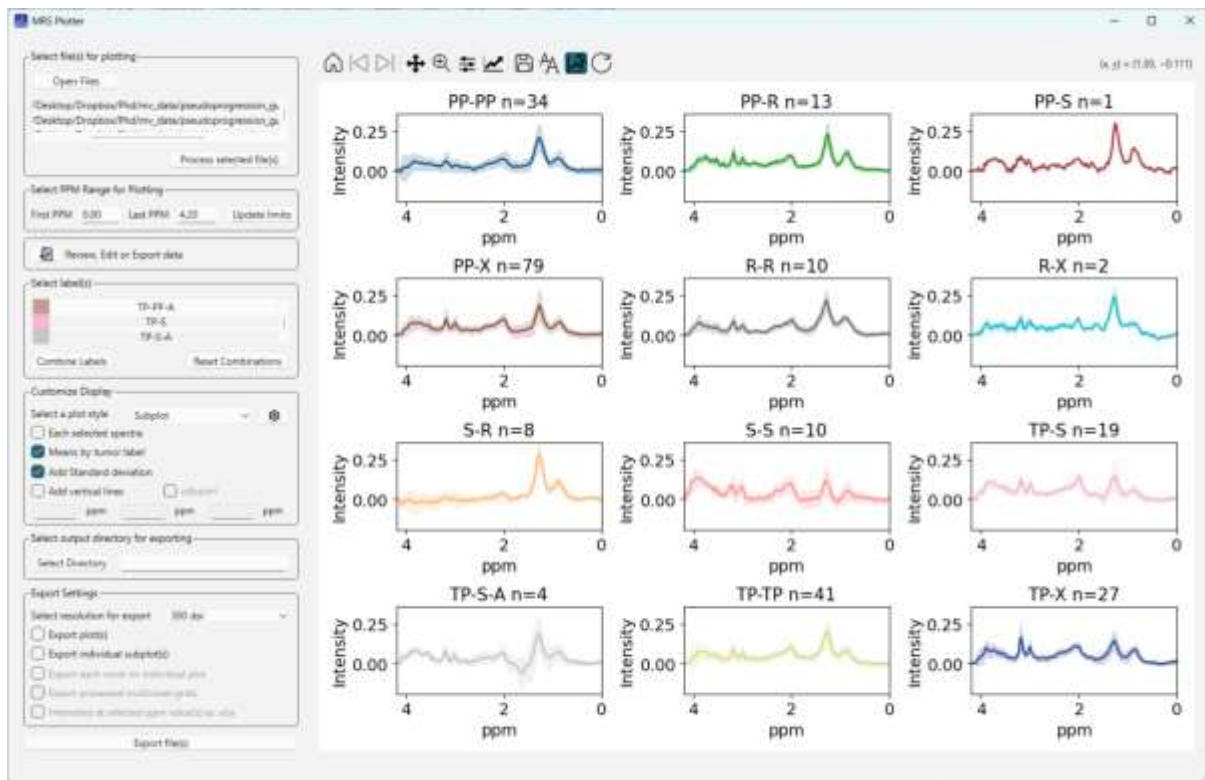


Figure 30 Labels from the pseudo progression multi voxel dataset with legends removed.

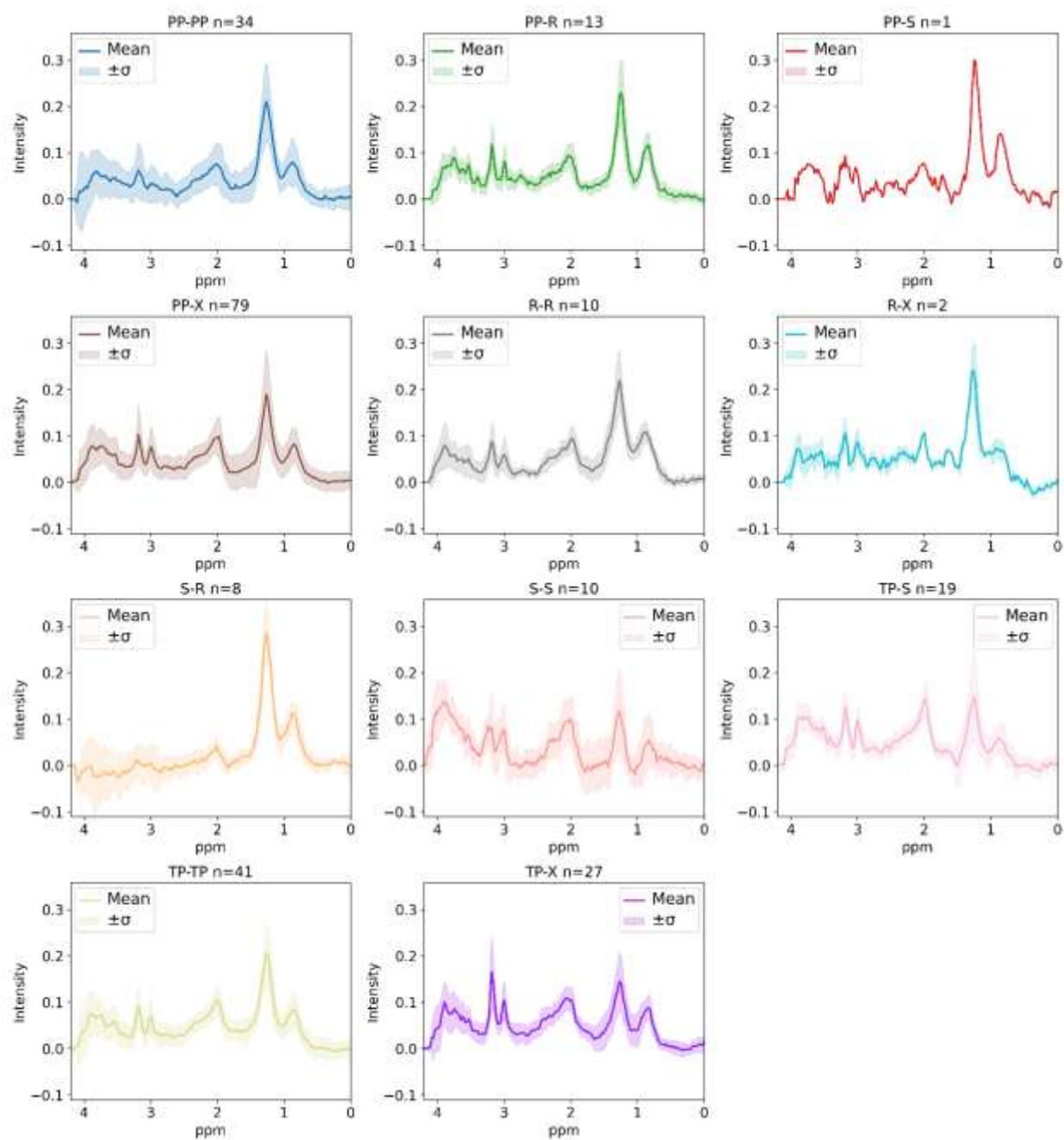


Figure 31. The previous figure when exported.

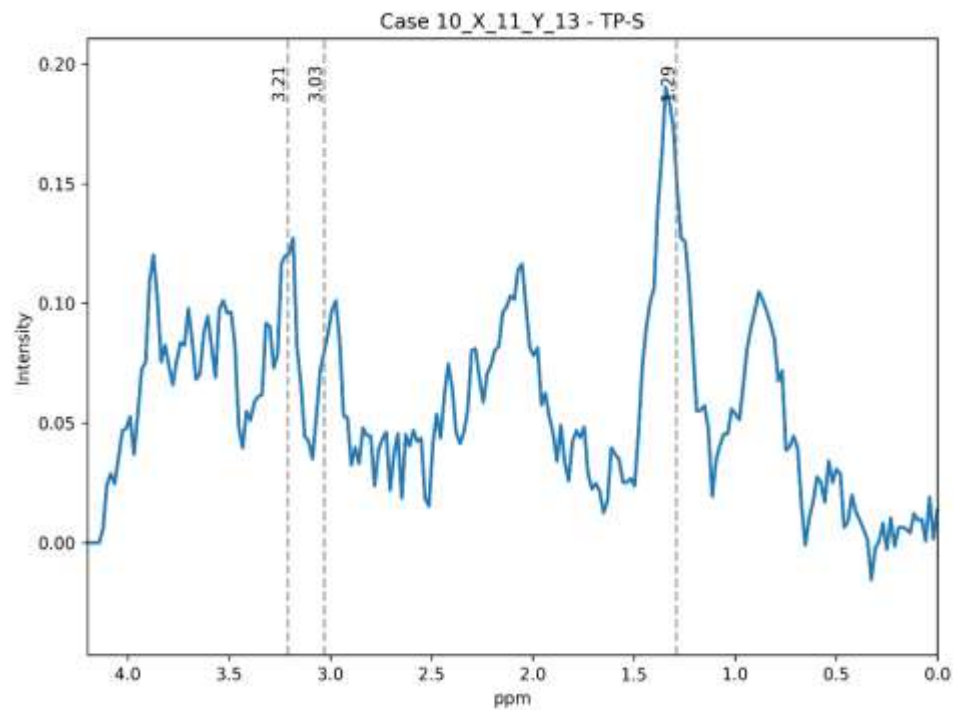


Figure 32. Exported spectra of a single voxel of a multi voxel case. The x and y positions are noted in the title and in the same of the exported file.

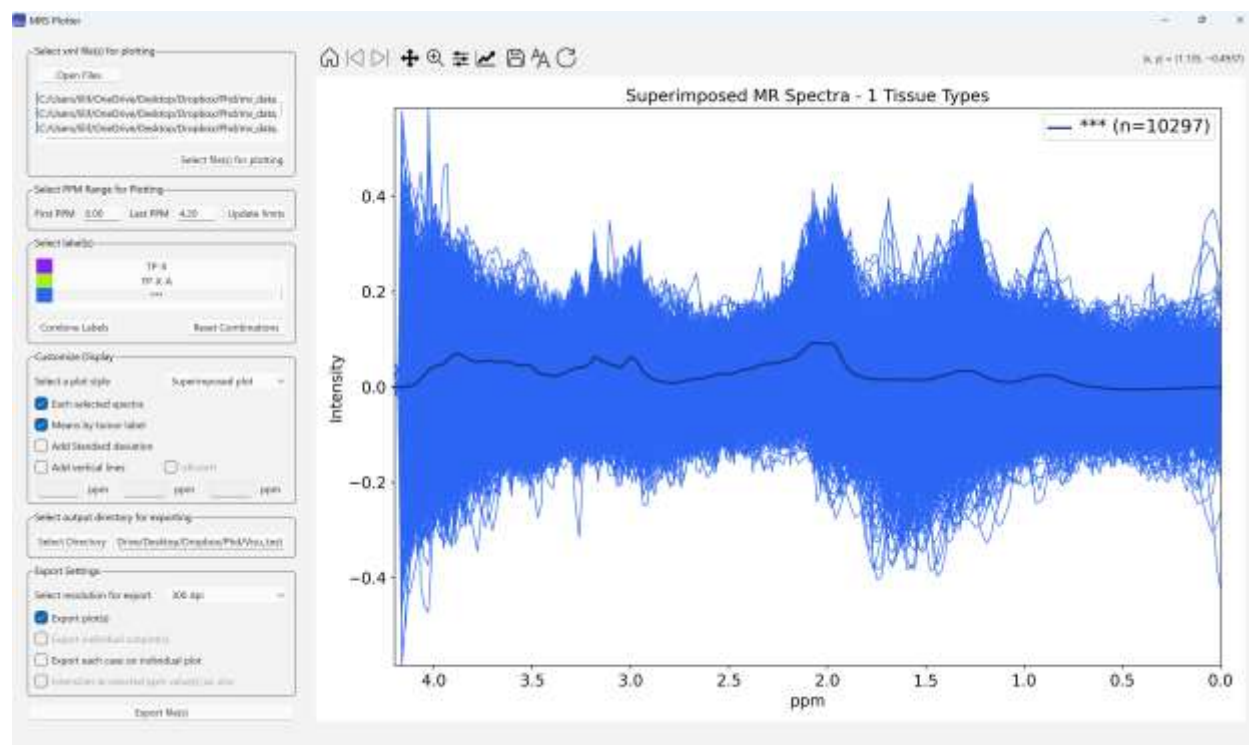


Figure 33. 10.297 '***' labelled spectra from the pseudo progression dataset (mean with dark blue)

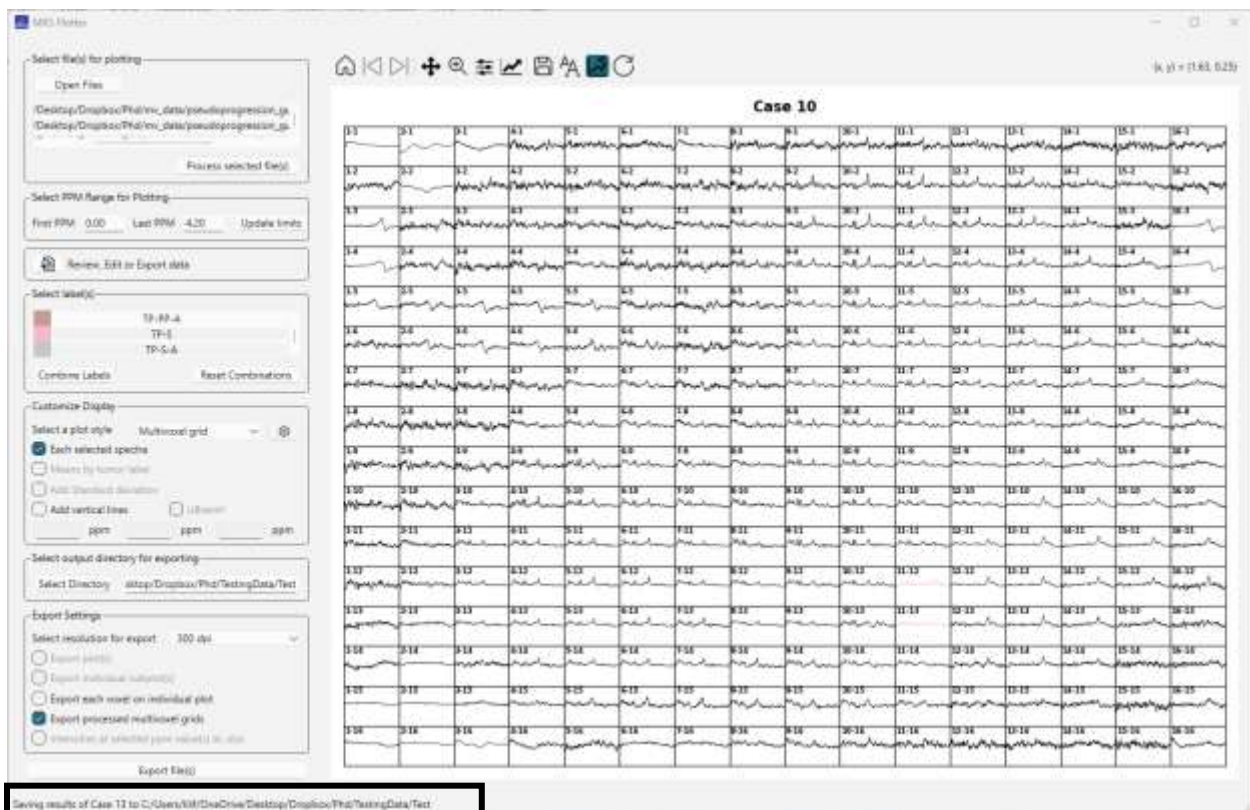


Figure 34 When multivoxel grid plot type is selected only one case is shown. But as can be seen on the status bar (blue rectangle) all processed cases can be exported.

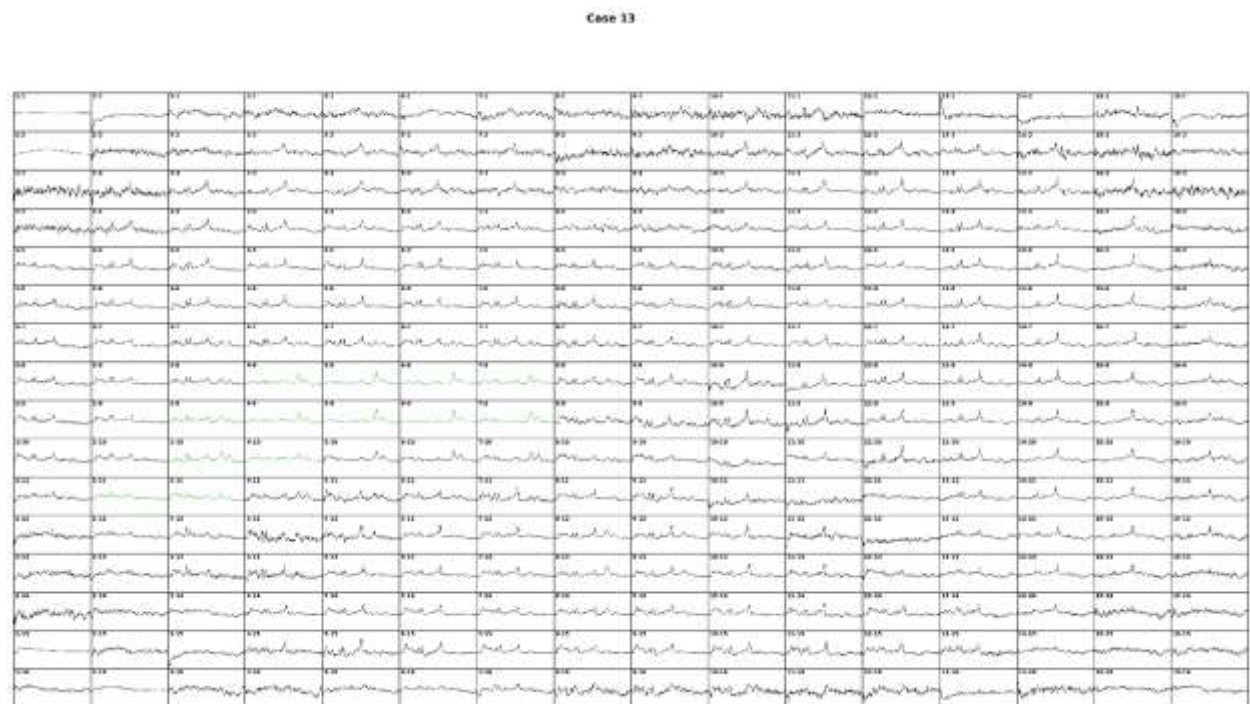


Figure 35 Grid view of Case 13. PP-R voxels are shown with green.

14. References

1. Mocioiu V, Ortega-Martorell S, Olier I, Jablonski M, Starcukova J, Lisboa P, Arús C, Julià-Sapé M. From raw data to data-analysis for magnetic resonance spectroscopy – the missing link: jMRUI2XML. *BMC Bioinformatics*. 2015;16(1):378. doi:10.1186/s12859-015-0796-5
2. Mocioiu V. JMRUI2XML plugin for jMRUI (Software).
3. Ortega-Martorell S, Olier I, Julià-Sapé M, Arús C. SpectraClassifier 1.0: a user friendly, automated MRS-based classifier-development system. *BMC Bioinformatics*. 2010;11(1):106. doi:10.1186/1471-2105-11-106
4. Stefan D, Cesare FD, Andrasescu A, Popa E, Lazariev A, Vescovo E, Strbak O, Williams S, Starcuk Z, Cabanas M, Ormond D van, Graveron-Demilly D. Quantitation of magnetic resonance spectroscopy signals: the jMRUI software package. *Meas Sci Technol*. 2009;20(10):104035. doi:10.1088/0957-0233/20/10/104035
5. Tate AR, Underwood J, Acosta DM, Julià-Sapé M, Majós C, Moreno-Torres À, Howe FA, van der Graaf M, Lefournier V, Murphy MM, Loosemore A, Ladroue C, Wesseling P, Luc Bosson J, Cabañas ME, Simonetti AW, Gajewicz W, Calvar J, Capdevila A, Wilkins PR, Bell BA, Rémy C, Heerschap A, Watson D, Griffiths JR, Arús C. Development of a decision support system for diagnosis and grading of brain tumours using in vivo magnetic resonance single voxel spectra. *NMR in Biomedicine*. 2006;19(4):411-434. doi:10.1002/nbm.1016
6. Early pseudoprogression and progression lesions in glioblastoma patients are both metabolically heterogeneous - Ungan - 2024 - NMR in Biomedicine - Wiley Online Library. Accessed March 20, 2025. <https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/nbm.5095>